

ACK

COUNTRY STILL RUNS MUCH OF ITS DIGITAL FUTURE
NOMY? BY **PALAK AGARWAL**

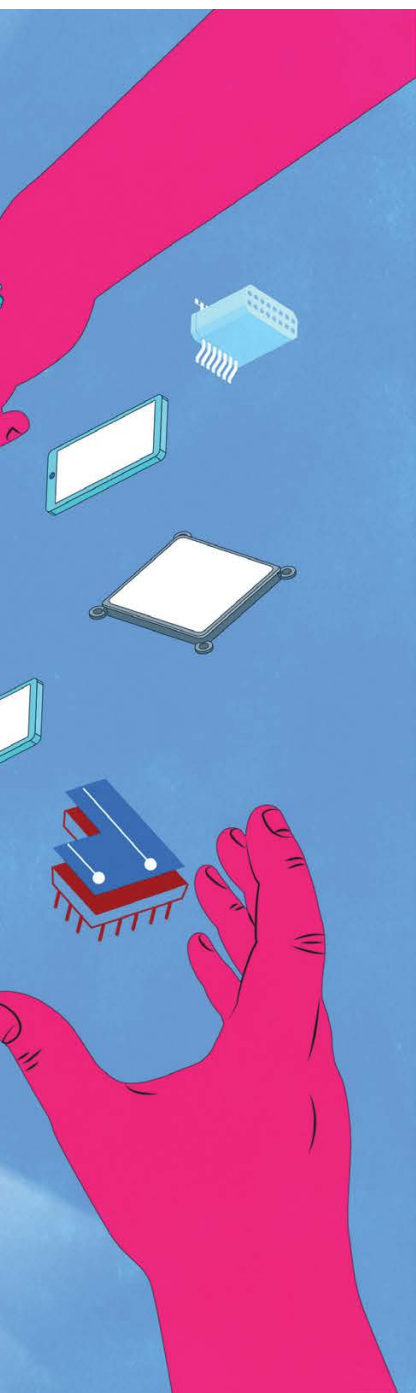
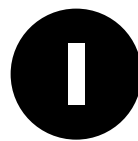


ILLUSTRATION BY **RAJ VERMA**

INDIA'S MISSING LAYER

- ▶ **India has become the world's second-largest smartphone manufacturer and the largest exporter of smartphones to the US. Yet 70-80% electronic components and 90-95% of its semiconductors are imported, limiting domestic value addition**
- ▶ **India remains overwhelmingly dependent on imported chips, especially CPUs and advanced semiconductors**
- ▶ **India generates nearly 20% of the world's data and has built a \$300-billion IT industry, yet stores only about 3% of global data**
- ▶ **The rise of AI has exposed gaps in India's technology stack, particularly in chips, GPUs and foundational models.**
- ▶ **India's future technology ambitions depend on building intellectual property, semiconductor capabilities and AI infrastructure**



INDIA EXPORTS SMARTPHONES

to America, writes software for the world and generates one-fifth of global data. Yet, the chips powering its devices, the Graphics Processing Units (GPUs) training its AI models and much of the software running its enterprises are controlled by foreign companies.

From hardware to software, much of India's technology stack is built on technologies developed elsewhere. This despite a population of over 1.4 billion—the largest in the world—and a deep pool of world-class talent.

Experts warn that India's reliance on foreign technologies mirrors a modern form of “colonialism” where unseen chains of dependency continue to tighten.

THE BATTLE OF OWNERSHIP

India has created a \$300-billion IT industry by exporting software to the world and is rapidly scaling electronics manufacturing. Over 99% of all mobile phones sold in India are now assembled domestically. Yet, beneath this success lies a striking dependence.

As per government data, India is now the world's second-largest mobile phone manufacturer, producing over 330 million handsets annually across more than 300 manufacturing facilities. Exports have grown even faster, surging 127-fold from ₹1,500 crore in FY15 to ₹2 lakh crore in FY25. Apple alone exported devices worth a record ₹1.1 lakh crore from India in 2024, marking a 42% year-on-year increase.

However, critical layers of the ICT ecosystem, right from semiconductors and electronic components to cloud infrastructure, enterprise software, and advanced computing remain largely imported or controlled by global technology giants.

Yes, in an interconnected world, global trade is built on interdependence. But disruptions of recent years—from the pandemic and supply-chain shocks to geopolitical tensions, the most recent being the war between the US and Iran, and export controls—have exposed the risks of excessive dependence and reignited the push for greater self-reliance.

This raises a fundamental question: Why are we so dependent on imports despite scale, talent pool, and technological ambitions?

“Until a decade ago, globalisation was the dominant narrative. But Covid-19, geopolitical tensions, supply-chain disruptions, and strategic realignments have exposed the risks of excessive